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Adaptive multi-objective Bayesian optimization approach for capacity planning of the interconnected offshore wind farms

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ABSTRACT

The concept of interconnected offshore wind farms (OWFs) presents a promising future that motivates this paper to develop a collaborative capacity planning model covering the system's economy, sustainability, and security needs. The model is optimized by an improved adaptive multi-objective Bayesian optimization (AMBO) algorithm, which seeks the Pareto optimal capacity allocation schemes of OWFs, offshore cables, battery energy storage systems, and hydrogen electrolysis units to form a Pareto front. Compared with other commonly used multi-objective optimization algorithms, the AMBO demonstrates its superiority in generating a more informative Pareto front sample-efficiently. In addition, it gets rid of the need for parameter tuning, thereby avoiding the introduction of planner subjectivity into the planning. The proposed planning model accounts for the inevitable simulation deviation resulting from the inherent variability of RES by modeling it into a noise function, whose side effects on the optimization can be mitigated through a probabilistic surrogate model. This approach ensures accurate performance evaluation of capacity allocation schemes without extensive simulations. With the aforementioned methodologies, the potential of OWF interconnection and integrated green hydrogen production to increase renewable power consumption and fortify the stability of power systems is demonstrated in the study.

1. Introduction

Renewable energy sources (RESs) have become pivotal when governments worldwide commit to net-zero targets [1,2]. Among these, offshore wind energy stands out as a key player. In 2024, the offshore wind market witnessed substantial growth, with 8 GW of new capacity commissioned globally. This impressive addition brought the total global offshore wind capacity to 83.2 GW, marking a significant milestone [3].

Considering the thriving growth of offshore wind farms (OWFs), the concept of interconnecting OWFs becomes a transformative development in the utilization of renewable energy, offering a multitude of potential benefits. By creating a networked offshore grid system, energy can be redistributed efficiently across different locations. This interconnection allows for a more stable energy supply, capable of adapting to fluctuating demands and reducing the reliance on traditional energy sources. Additionally, the interconnected OWFs' framework offers a perfect platform for the integration of hydrogen production, which not only provides a profitable supply of green hydrogen but also utilizes surplus wind power and reduces wind curtailment [4].

The burgeoning expansion of OWFs, coupled with their potential for interconnection and green hydrogen production, marks a significant leap toward a sustainable energy future. Despite its promise, the coordinated planning of interconnected OWFs integrated with hydrogen production remains underexplored. Developing such complex infrastructure requires addressing economic, environmental, and secure dimensions simultaneously. Therefore, a multi-objective optimization (MOO) approach is essential to provide planners with objective solutions that capture the trade-offs between conflicting goals.

1.1. Literature review and research gaps

Contemporary research on OWFs spans a broad spectrum, from technological advancements in equipment to analyses of OWF characteristics and comprehensive financial assessments. For instance, the design of inter-farm collector systems is optimized in [5,6] to achieve a higher transmission efficiency with less investment in cables. Additionally, planning models of OWFs integrated with other energy forms have been proposed, including tidal power [7], natural gas [8], hydrogen-ammonia systems [9], and battery energy storage systems [10], where a sequential "planning + operational" framework is employed to determine optimal storage capacity under high-impact, low-probability

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events, effectively linking OWF topology design with resilience enhancement. Financial analyses of OWFs also cover various dimensions, such as over-planting rates [11], wind uncertainty [12], and AC-DC transmission options [13], which demonstrates a recent decrease in investment costs for OWFs and associated cabling. Such a trend has brought projects like the *North Sea Wind Power Hub* into practice, aiming to interconnect OWFs in different locations. However, integrated and collaborative planning of OWFs, hydrogen systems, and offshore cables while considering multidimensional needs remains limited.

The needs in the power system planning normally cover the quests for economy, sustainability, and security, which often lead to a trade-off, as achieving all three simultaneously is extremely challenging. Traditional models use a weighted sum method to merge various factors like investment [14–16], load shedding [14,16], RES power curtailment [14], and carbon tax [15] into one objective. This approach, though straightforward, can miss complex trade-offs and be subjective, as planners choose the weights based on personal judgment.

On the other hand, MOO offers a powerful framework for addressing the inherent trade-offs in power system planning by generating a set of Pareto-optimal solutions rather than a single compromise. This approach allows planners to evaluate a spectrum of options, facilitating more informed decision-making through a combination of quantitative insights and expert judgment [17]. MOO has been increasingly applied to offshore energy systems. For instance, Neshat et al. [18] proposed a multi-objective swarm optimization method for improving the performance of hybrid wave-wind systems. Similarly, NSGA-II has been employed in [19] to optimize investment and operational efficiency in offshore multi-energy systems, and in [20], the algorithm is delicately enhanced with an AI-model to design blades of offshore wind turbines with reduced mass and deflection.

Despite their popularity, existing MOO algorithms face key limitations when applied to complex offshore network planning involving interconnected OWFs. These algorithms can be broadly categorized into two groups. The first group includes adaptive weighting methods, such as the adaptive weighted-sum algorithm [21] and its improved variant [22], which reduce subjectivity but often lead to uneven Pareto front distributions and require manual parameter tuning. The second group consists of heuristic search-based algorithms, including DMOGWA [18], NSGA-II [19], and MOAVOA [23], which are effective for nonlinear problems but highly sensitive to population size, leading to inconsistent performance [24]. Additionally, these methods are computationally intensive for large-scale problems and often suffer from overlapping or poorly distributed Pareto solutions [22]. Crucially, neither category adequately addresses Pareto front diversity, a vital aspect for supporting flexible and balanced planning decisions in real-world systems.

Except for the unsatisfactory performance of current MOO algorithms, another major challenge in OWF planning is deviation in operation simulation of the system due to variability in RES generation. Traditional approaches confront an unavoidable challenge, necessitating a compromise between objectivity and computational efficiency. To elaborate, capacity planning models are crucial for preparing power systems to better meet future demands. However, the scenarios these systems will actually encounter are unpredictable, leading to deviations between computer simulations and real-world performance. This is especially true for the systems with high penetration of RESs, whose output power fluctuates unpredictably. Traditional approaches rely on *typical scenarios* [25] or *extending dispatch cycles* [26] to lessen this variability but inevitably compromise either objectivity or computational efficiency. To overcome these shortcomings, this study introduces a noise-modeling approach that incorporates inevitable deviation between the simulation environment and real-world environment into the planning model as a noise function, offering a more robust response to the unpredictable and boundless nature of RES scenarios.

1.2. Summary of major contributions

To address the identified challenges, this study proposes a collaborative capacity planning framework that jointly optimizes the capacity allocation for OWFs, offshore transmission cables, hydrogen electrolyzers, and BESSs. The planning model is designed to unlock the full potential of interconnected OWFs integrated with hydrogen production, providing a practical solution to spatial and temporal mismatches between renewable generation and load demand. This framework is powered by a newly developed AMBO algorithm, which improves upon existing MOO approaches by efficiently generating a diverse and well-distributed Pareto front without requiring manual parameter tuning. Furthermore, by incorporating a noise modeling mechanism that accounts for the variability and uncertainty in RES output, the proposed method enhances robustness in planning outcomes without the computational burden of scenario-based simulations. Collectively, these advancements address key limitations in the literature and provide planners with a reliable and scalable approach to support long-term offshore energy development across economy, sustainability, and security dimensions.

Compared with existing works, three main contributions of the article are summarized as follows:

(i) The study improves the existing Bayesian-based MOO algorithms by proposing the AMBO algorithm, an enhanced MOO algorithm applied to planning problems *for the first time*. This approach generates a more informative Pareto front sample-efficiently, without introducing predetermined parameters that could introduce planner bias.
(ii) A multi-objective capacity planning model that leverages the benefits of interconnected OWFs and hydrogen production integration is proposed in the study. The spatial and temporal mismatches between RES power and load are well addressed and the $N-1$ contingency analysis is incorporated in the planning for a more secure grid.
(iii) Compared with the existing planning methods, the deviation between the simulation and real-world environment is directly modeled into a noise function. Through the probabilistic surrogate model within AMBO, the side effects of noise in misleading the optimization can be properly mitigated, enabling the attainment of a well-performed capacity allocation scheme without extensive computational resources.

The rest of the article is organized as follows: The motivation and prerequisite knowledge of the proposed approach are introduced in Section 2. The formulation of the capacity planning model is presented in Section 3. The mathematical details of AMBO are presented in Section 4. Case study and conclusions are provided in Sections 5 and 6.

2. Problem formation

This section introduces the motivation of interconnected OWFs and provides detailed explanations about the source of simulation deviation along with its solution.

2.1. The motivation of OWF interconnection

The development of OWFs offers a strategic avenue for enhancing transmission capacity, which in turn benefits the power grid. Many regions worldwide face a geographical dichotomy: their load centers are far away from places possessing abundant wind resources. This spatial mismatch and the temporal mismatch between load demands and the uncertain OWF output strain the existing onshore transmission infrastructure, leading to RES power curtailment and load shedding happening at the same time. Moreover, the scarcity of land coupled

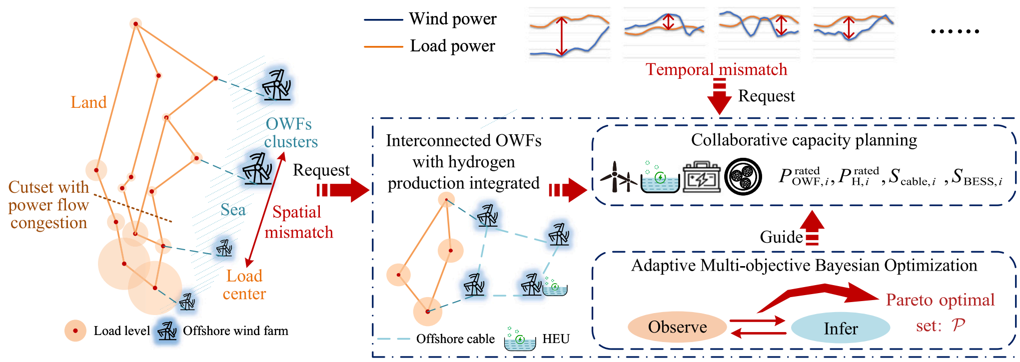


Fig. 1. The motivation for planning the interconnected OWFs integrated with hydrogen production.

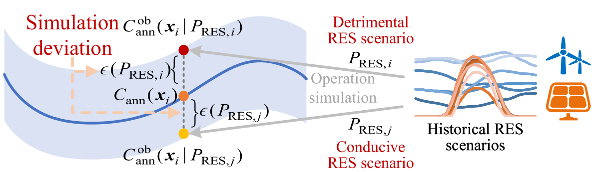


Fig. 2. The influence rendered by the simulation deviation caused by the RES variability on the annual cost.

with environmental considerations constrains the expansion of onshore transmission lines. Thus, the interconnection of OWFs presents a viable solution to balance the provincial load distribution, mitigating grid stress and fostering a more reliable energy supply, as depicted in Fig. 1. However, as a huge project to invest in, the planning has to cover the power system's comprehensive requirements and properly tackle the inherent simulation deviation.

2.2. Relationship between the noise and deviation of operation simulation

Unlike the security and sustainability performance that can be evaluated under extreme situations with fixed RES scenarios, the economy evaluation of a capacity allocation scheme depends on the simulation of a power system's normal operation. The future RES scenarios that confront a power system to be planned have inherent unpredictability and boundlessness, introducing an unavoidable deviation between the simulation and real-world environment in the evaluation. The existence of such deviation indicates the real economic performance of a power system can hardly be precisely evaluated. Traditional planning models utilize the typical scenarios or extension of simulation cycles to diminish such deviations. However, these methods rely on historical RES scenarios, which, while insightful, cannot perfectly replicate the complex future. Essentially, the simulation deviation can be seen as a noise function added to the real objective value, as shown in (1):

$$C_{\text{ann}}^{\text{ob}}(\boldsymbol{x} \mid P_{\text{RES}}) = C_{\text{ann}}(\boldsymbol{x}) + \epsilon(P_{\text{RES}}), \tag{1}$$

where C_{ann} is the economy objective and $\boldsymbol{x}$ is the capacity allocation scheme, their specific meanings will be explained in the next section; $C_{\text{ann}}^{\text{ob}}$ is the outcome that can be attained through computer simulation; P_{RES} represents the output power of all the RESs in the power system; $\epsilon(P_{\text{RES}})$ is the noise term representing the simulation deviation. In a power system with high penetration of RES power, the simulation deviation is mainly attributed to the variability of RES, but it is also introduced by the uncertainty of load, the inaccurate estimate or shifting of grid parameters, and many other uncertain aspects. These factors have relatively smaller influences. As a result, only the variability of RES is concentrated and further analyzed in the study.

Their relationship is depicted in Fig. 2. When simulations are conducted under a conducive RES scenario, characterized by abundant and stable power output from RESs, the simulated cost is typically lower than what might be realized in practice. Conversely, detrimental RES scenarios can lead to security issues and more expensive operations. This deviation is captured by the noise term in the model. Whereas the noise term will inevitably mislead the optimization process, the proposed AMBO algorithm is proven to be able to effectively address this issue, ensuring that the capacity allocation scheme developed by the planning model performs well in real operational conditions.

3. Scheme evaluation model

3.1. Overall structure

The scheme evaluation model comprises two distinct sub-problems: the economic operation and the security correction. Within these sub-problems, three carefully selected objectives address the core pillars of modern power system planning: economy, sustainability, and security. The first two objectives, annual cost and RES consumption, are calculated by the optimization of the economic operation (sub-problem 1), reflecting economic feasibility and environmental sustainability, respectively. The annual cost accounts for all key system expenditures and potential income sources, ensuring economic viability, while the RES consumption objective aligns with national decarbonization targets and reflects the system's ability to integrate green energy. For assessing RES accommodation capability, OWFs are configured to operate at maximum output, while annual cost is evaluated using random RES scenarios drawn from historical or generated datasets. As introduced in Section 2.2, this evaluation introduces a noise term that must be effectively managed by the AMBO algorithm.

The third objective, the reserved-power-to-investment ratio (RPIR), emerges from the security correction (sub-problem 2) and incorporates $N-1$ contingency analysis for both existing transmission lines and offshore cables, ensuring system reliability while balancing redundancy against economic practicality. The structure and simulation process of the evaluation model is illustrated in Fig. 3.

These three objectives – annual cost, RES consumption, and RPIR – are interrelated in complex and often conflicting ways. Improvements in one dimension can lead to diminishing returns or adverse effects in another, depending on how investment is distributed across system components. For instance, while increased renewable capacity can reduce operational costs initially, excessive investment ultimately yields limited additional sustainability benefits at a steep economic price. Similarly, certain instances reveal that enhancing system security through redundancy comes at the expense of RES integration or affordability. In some cases, resource allocation decisions create direct trade-offs between sustainability and reliability, while in others, careful balancing allows synergistic improvements. These nuanced and case-dependent relationships highlight that no single objective can adequately capture the multi-faceted goals of offshore system planning. As

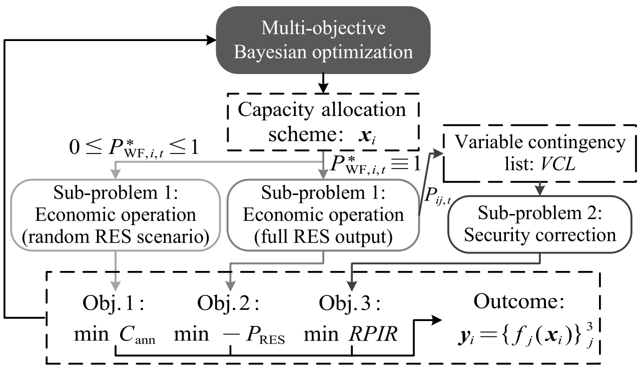


Fig. 3. The structure of scheme evaluation model.

such, employing a multi-objective optimization framework is essential for transparently navigating these trade-offs and generating a diverse set of Pareto-optimal solutions, from which planners can select based on system-specific priorities and constraints.

3.2. The evaluation of economic and sustainable objective functions

3.2.1. Economy objective functions

The economy objective function, the system's annual cost, considers the equivalent annual investment, maintenance and operation cost, denoted as follows:

$$\arg\min_{\boldsymbol{x}} \; C_{\mathrm{ann}} = \overbrace{\sum_{k\in\Omega} CPR_k C_k}^{\text{investment}} + \overbrace{\sum_{k\in\Omega} \kappa_k C_k}^{\text{maintenance cost}} + \overbrace{(C_{\mathrm{g}} - R_{\mathrm{H}})}^{\text{operation cost}}, \tag{2}$$

with

$$\boldsymbol{x} = \{[P^{\mathrm{rated}}_{\mathrm{OWF},i}]_1^{N_{\mathrm{OWF}}}, [P^{\mathrm{rated}}_{\mathrm{H},i}]_1^{N_{\mathrm{H}}}, [S_{\mathrm{cable},i}]_1^{N_{\mathrm{cable}}}, [S_{\mathrm{BESS},i}]_1^{N_{\mathrm{BESS}}}\}, \tag{3}$$

$$CPR_k = \frac{\tau(1+\tau)^{T_{\mathrm{life},k}}}{(1+\tau)^{T_{\mathrm{life},k}} - 1}, \quad k \in \Omega, \tag{4}$$

$\Omega = \{\mathrm{OWF}, \mathrm{H}, \mathrm{cable}, \mathrm{pipe}, \mathrm{conv}, \mathrm{BESS}\}$,

where C_{ann} is the annual cost; N_{OWF}, N_{cable}, N_{H}, N_{BESS} are the total number of OWFs, offshore cables, hydrogen electrolyzers and BESSs respectively; $P^{\mathrm{rated}}_{\mathrm{OWF},i}$ and $P^{\mathrm{rated}}_{\mathrm{H},i}$ are the rated power of i^{th} OWF and hydrogen electrolyzer; $S_{\mathrm{cable},i}$ and $S_{\mathrm{BESS},i}$ are the capacities of i^{th} offshore cable and batter energy storage system (BESS); CPR_k, C_k are the capital recovery factor and total investment of equipment k; κ_k is the operating expense ratio of equipment k; C_{g} is the annual generation cost; R_{H} is the annual hydrogen sale revenue; $T_{\mathrm{life},k}$ is the designed lifetime of equipment k; τ is the discount rate; Ω is the equipment set comprising components that are planned in the model, including OWFs, hydrogen electrolyzers, cables, hydrogen pipes, converters, and BESSs. While BESS is included in this study to capture basic charging and discharging operations for energy balancing and backup, it is not the central focus of the proposed planning framework, which primarily investigates how offshore wind interconnection and hydrogen integration contribute to system sustainability and security. Readers interested in advanced BESS applications, such as resilience-oriented sizing [10] or wind–wave coordinated operation [27], are referred to relevant studies.

Equations for investment cost C_k of each type of equipment are concluded from [28–30] and detailed in (5)–(8).

$$C_{\mathrm{OWF}} = \sum_i^{N_{\mathrm{OWF}}} C_{\mathrm{OWF},i} = \sum_i^{N_{\mathrm{OWF}}} \Big[\overbrace{\alpha_{\mathrm{OWF}} (P^{\mathrm{rated}}_{\mathrm{OWF},i})^{\gamma_{\mathrm{OWF}}}}^{\text{turbine cost}} + \overbrace{\beta_{\mathrm{OWF}} (P^{\mathrm{rated}}_{\mathrm{OWF},i})^{\delta_{\mathrm{OWF}}}}^{\text{foundation cost}}$$
$$+ \overbrace{\alpha_{\mathrm{plat}} P^{\mathrm{rated}}_{\mathrm{OWF},i} + \beta_{\mathrm{plat}}}^{\text{platform cost}} + \overbrace{\alpha_{\mathrm{conv}} P^{\mathrm{rated}}_{\mathrm{OWF},i}}^{\text{converter cost}} \Big], \tag{5}$$

$$C_{\mathrm{cable}} = \sum_i^{N_{\mathrm{cable}}} C_{\mathrm{cable},i} = \sum_i^{N_{\mathrm{cable}}} \left[\alpha_{\mathrm{cable}} \exp\Big(\frac{\gamma_{\mathrm{cable}} S_{\mathrm{cable},i}}{100}\Big) + \beta_{\mathrm{cable}} \right] L_{\mathrm{cable},i}, \tag{6}$$

$$C_{\mathrm{H}} = \sum_i^{N_{\mathrm{H}}} C_{\mathrm{H},i} = \sum_i^{N_{\mathrm{H}}} \Big[\overbrace{\alpha_{\mathrm{H}} (P^{\mathrm{rated}}_{\mathrm{H},i})^{\beta_{\mathrm{H}}}}^{\text{electrolysis cost}} + \overbrace{\alpha_{\mathrm{pipe}} L_{\mathrm{pipe}} P^{\mathrm{rated}}_{\mathrm{H},t} \eta_{\mathrm{H}} / \mathrm{LHV}_{\mathrm{H}}}^{\text{pipe cost}} + \overbrace{\alpha_{\mathrm{Hconv}} P^{\mathrm{rated}}_{\mathrm{H},i}}^{\text{converter cost}} \Big], \tag{7}$$

$$C_{\mathrm{BESS}} = \sum_i^{N_{\mathrm{BESS}}} C_{\mathrm{BESS},i} = \sum_i^{N_{\mathrm{BESS}}} \left[\alpha_{\mathrm{BESS},1} S_{\mathrm{BESS},i} + \alpha_{\mathrm{BESS},2} P^{\max}_{\mathrm{BESS},i} \right], \tag{8}$$

where $L_{\mathrm{cable},i}$ is the length of i^{th} offshore cable, and L_{pipe} is the total length of hydrogen pipes; α_k, β_k, γ_k, and δ_k are constants; η_{H} and $\mathrm{LHV}_{\mathrm{H}}$ are the efficiency of hydrogen electrolyzer and lower heating value of hydrogen; $P^{\max}_{\mathrm{BESS,i}}$ is the maximum output power of i^{th} BESS.

The calculation of operation cost is shown in (9)–(10).

$$C_{\mathrm{g}} = T_{\mathrm{d}} \sum_i^{N_{\mathrm{g}}} \sum_t^{T} (\mathrm{a}_i P^2_{\mathrm{g},i,t} + \mathrm{b}_i P_{\mathrm{g},i,t} + \mathrm{c}_i), \tag{9}$$

$$R_{\mathrm{H}} = T_{\mathrm{d}} \sum_i^{N_{\mathrm{H}}} \sum_t^{T} p_{\mathrm{H}} H_{i,t}, \tag{10}$$

where T_{d} and T are the total number of dispatch cycles in one year and the number of dispatch intervals in one dispatch cycle, which, in this case, are 365 and 24 respectively; N_{g} is the total number of generators; $P_{\mathrm{g},i,t}$ is the output power of i^{th} generator at time t; a_i, b_i, c_i are fuel cost coefficients of i^{th} generator; p_{H} is the price of green hydrogen, and $H_{i,t}$ is the weight of the hydrogen produced by i^{th} hydrogen electrolyzer at time t. A higher hydrogen price p_{H} makes each unit of hydrogen more profitable, thereby increasing the economic value of utilizing surplus renewable energy through electrolyzers and incentivizing greater deployment of hydrogen production capacity in the model.

3.2.2. Sustainability objective function

The sustainability objective is the total RES power consumed by the power system. Since OWFs are the concentration of this study, only the RES power from OWFs is considered but the model is compatible for the admission of other kinds of renewable power.

$$\arg\min_{\boldsymbol{x}} \; -P_{\mathrm{RES}} = -\sum_i^{N_{\mathrm{OWF}}} \sum_t^{T} P_{\mathrm{OWF},i,t} \tag{11}$$

where $P_{\mathrm{OWF},i,t}$ is the consumed power of i^{th} OWF at time t.

3.3. Sub-problem 1: Economic operation

Sub-problem 1 is solved twice with different RES output in (16) to evaluate a scheme. When the RES output in (16) is set according to a random RES scenario, its outcome defined in (9) is used to calculate C_{ann} in (2). When all the OWFs are at their maximum output, sub-problem 1 is solved to get P_{RES} in (11), and the acquired power flow is used to calculate the variable contingency list (VCL) used in sub-problem 2.

3.3.1. Objective function

The purpose of the economic operation is to minimize the generation cost C_{g} previously defined in (9).

$$\arg\min_{\boldsymbol{P}} \; C_{\mathrm{g}}, \tag{12}$$

with

$$\boldsymbol{P} = [P_{\mathrm{g},i,t}, P_{\mathrm{OWF},i,t}, P_{\mathrm{H},i,t}, P_{\mathrm{BESS},i,t}], \tag{13}$$

where $P_{\mathrm{H},i,t}$ is the consumed power of i^{th} electrolysis at time t; $P_{\mathrm{BESS},i,t}$ is the output or input power of i^{th} BESS at time t, depending on its sign.

3.3.2. The constraints of traditional generator

$$-\delta_g P_{g,i}^{max} \le P_{g,i,t+1} - P_{g,i,t} \le \delta_g P_{g,i}^{max}, \tag{14}$$

$$P_{g,i}^{min} \le P_{g,i,t} \le P_{g,i}^{max}, \tag{15}$$

where δ_g is the ramp rate of traditional generators; $P_{g,i}^{max}$, $P_{g,i}^{min}$ are the maximum and minimum output power of i^{th} generator.

3.3.3. The constraints of OWF

$$0 \le P_{OWF,i,t} \le P_{OWF,i,t}^{max} = P_{OWF,i}^{rated} P_{OWF,i,t}^{*}, \tag{16}$$

where $P_{OWF,i,t}^{*}$ is the per unit values of maximum offshore wind power, dependent on the weather condition. In the evaluation of annual cost, these values are randomly chosen from historical data or generated RES scenarios, while $P_{OWF,i,t}^{*}$ is set to equal to one when evaluating RES accommodation capability. Moreover, the inequalities in (16) imply that the curtailment of RES power is permitted in the simulation.

3.3.4. The constraints of hydrogen electrolyzer

Despite being situated on OWF platforms, the process of offshore hydrogen production is largely similar to that on land. The energy conversion process from electricity to hydrogen is modeled as introduced in [31].

$$H_{i,t} = \eta_H \frac{P_{H,i,t}}{LHV_H}, \tag{17}$$

$$0 \le P_{H,i,t} \le P_{H,i}^{rated}. \tag{18}$$

3.3.5. The constraints of BESS

$$\begin{cases} -P_{BESS,i}^{max} \le P_{BESS,i,t} \le P_{BESS,i}^{max} \\ S^{min} \le S_{i,t} \le S_{BESS,i} \end{cases}, \tag{19}$$

$$\begin{cases} S_{i,t+1} = S_{i,t} - \lambda_c P_{BESS,i,t} \Delta t, & P_{BESS,i,t} \le 0 \\ S_{i,t+1} = S_{i,t} - \dfrac{P_{BESS,i,t}}{\lambda_d} \Delta t, & P_{BESS,i,t} \ge 0 \\ S_{i,0} = S_{i,T} \end{cases}, \tag{20}$$

$$P_{BESS,i}^{max} = s S_{BESS,i}, \tag{21}$$

where $S_{i,t}$ is the reserved energy of i^{th} BESS at time t; S^{min} is the minimum allowed energy of BESS; λ_c, λ_d are the efficiency of charge and discharge; Δt is the time interval of the dispatch, which equals 1 h in the study; s is a fixed ratio of the maximum output power $P_{BESS,i}^{max}$ to the maximum energy storage $S_{BESS,i}$.

3.3.6. The constraints of DC power flow

The study utilizes the DC power flow model and omits reactive power considerations for simplicity. Notably, all the offshore cables are assumed to be high-voltage AC cables.

$$P_{l,t} = B_l(\theta_{i,t} - \theta_{j,t}), \tag{22}$$

$$\begin{cases} -P_l^{max} \le P_{l,t} \le P_l^{max} \\ -\dfrac{\pi}{2} \le \theta_{i,t} \le \dfrac{\pi}{2} \end{cases}, \tag{23}$$

where $P_{l,t}$ is the power flow of branch l at time t; B_l is the line admittance of branch l; $\theta_{i,t}$ is the voltage angle at bus i at time t.

3.3.7. The power balance constraint

In economic operation, the power balance constraint permits no load shedding.

$$P_{s,i,t} + \sum_i^{N_l} \Psi_i P_{l,t} = P_{load,i,t}, \quad s \in \{H, BESS, g\}, \tag{24}$$

where $P_{s,i,t}$ is the power consumed or generated power by a hydrogen electrolyzer, BESS or tradition generator at bus i at time t, and $P_{load,i,t}$ is the load power of bus i at time t; N_l is the number of branches; Ψ_i is the bus-branch incidence matrix.

3.4. The evaluation of security objective function

3.4.1. Security objective function

The security objective, defined as the reserved-power-to-investment ratio (RIPR) is a measure that balances the reserved power available during contingencies against the annual investment in the power system infrastructure, defined as follows:

$$\arg\min_{x} \; -RPIR = -\frac{\sum_m^{N_l} \sum_t^{T} \sum_i^{N_{bus}} (P_{load,i,t} - P_{loss,m,i,t})}{\sum_{k\in\Omega} CPR_k C_k}, \tag{25}$$

where $P_{loss,m,i,t}$ is the load shedding of bus i under outage of branch m at time t.

3.4.2. Variable contingency list

To simplify the computational complexity of $N-1$ contingency analysis, the approach proposed in [32] is adopted, determined by assessing the impact of a branch outage on the post-contingency flows of other branches. It specifically includes only those branches that, if of outage, would cause a significant overload on other branches. The calculations of VCL are shown below:

$$\Gamma_{l,m,t} = \frac{P_{l,m,t} - P_l^{max}}{P_l^{max}}, \tag{26}$$

$$\Phi_m = \left\{ l \in N_l \mid \max\{\Gamma_{l,m,t}\} \ge \nu_1 \cup \sum_t^{T} \frac{\Gamma_{l,m}^t}{T} \ge \nu_2 \right\}, \tag{27}$$

$$VCL = \{ m \in N_l \mid \Phi_m \ne \emptyset \}, \quad \forall l, m \in N_u, \quad \forall t \in T, \tag{28}$$

where $P_{l,m,t}$ is the power flow in branch l under outage of branch m at time t and its calculations can be found in [32]; $\Gamma_{l,m,t}$ is the magnitude of violation in the power flow of branch l under outage of branch m time t; Φ_m is the set of branches with violated post-contingency flows under outage of branch m; N_l is the number of all existing lines and the offshore cables to be planned.

The definition of Φ_m in (27) is to select branches with their peak power flow exceeding the heat maximum (ν_1% overload) or their average overload exceeding emergency capacity (ν_2% overload).

3.5. Sub-problem 2: Security correction

Sub-problem 2 is set to evaluate the minimum load shedding that a scheme can achieve when the lines in the VCL are of outage. It is solved to get $RPIR$ in (25).

3.5.1. Objective function

The objective of the security correction is the minimization of the total load shedding P_{loss}. Notably, $P_{loss,m,i,t} = 0$ if branch m is not included in the VCL.

$$\arg\min_{P'} \; P_{loss} = \sum_m^{M} \sum_t^{T} \sum_i^{N_{bus}} P_{loss,m,i,t}, \tag{29}$$

with

$$P' = [P_{g,m,i,t}, P_{OWF,m,i,t}, P_{H,m,i,t}, P_{BESS,m,i,t}], \tag{30}$$

where the subscription m means that the system is operating under the outage of line m.

3.5.2. Constraints

The security correction model, operating under $N-1$ contingency scenario adheres to the same constraints previously defined in (14)–(23) except the power balance constraint (24), which is adjusted as shown in (31). This modification allows for load shedding in contingency situations.

$$P_{\mathrm{s},m,i,t} + \sum_{i}^{N_l} \Psi_i P_{m,l,t} = P_{\mathrm{load},i,t} + P_{\mathrm{loss},m,i,t}. \tag{31}$$

4. Proposed adaptive multi-objective Bayesian optimization algorithm

As formerly introduced in Section 2.2, A key challenge in capacity planning is mitigating the impact of RES variability on the noisy outcomes $C_{\mathrm{ann}}^{\mathrm{ob}}(\boldsymbol{x})$ to accurately determine the actual annual cost $C_{\mathrm{ann}}(\boldsymbol{x})$.

To settle the problem the AMBO algorithm is proposed in the study improved from the methodology in [33]. The AMBO algorithm stands out from traditional MOO algorithms by directly optimizing the distribution of the Pareto fronts through the maximization of the hypervolume indicator, which is a measure widely employed to assess the quality of Pareto fronts.

In each iteration of the capacity planning model, the AMBO algorithm sends a capacity allocation scheme to the scheme evaluation model and receives the objectives values as outcomes. Then all the capacity allocation schemes and their three corresponding objective values accumulated in each iteration are used in the Gaussian process, a non-parametric model to infers a distribution over functions that best describes the relationship between capacities and each objective function.

The generated distribution is utilized to maximize the function *expected hypervolume improvement* to seek a candidate point to be evaluated, which is most likely to bring the maximum expected increase of hypervolume.

Although the Gaussian process model has inherent errors in its inference, it iteratively updates and improves its accuracy, using cumulative outcomes to converge to the true underlying functions. Thus, the optimal Pareto front can be approached without extensive evaluations of objective functions, which would otherwise be computationally expensive.

The detailed optimization process of AMBO is introduced in the following subsections.

4.1. Initialization

To initiate the process, a set of N_0 randomly chosen capacity allocation schemes, denoted as $\boldsymbol{X}_0$ is evaluated by the evaluation model. This brings an initial dataset $\boldsymbol{D}_0$ containing the observed objective function values $\boldsymbol{y}_0$, which include the three objectives specified in (2), (11) and (25). Specific definitions of $\boldsymbol{X}_0$ and $\boldsymbol{Y}_0$ are provided as follows:

$$\boldsymbol{D}^0 = \{\boldsymbol{X}^0, \boldsymbol{Y}^0\}, \tag{32}$$

where

$$\begin{aligned}\boldsymbol{X}^0 =& \{\boldsymbol{x}_i\}_1^{N_0} = \{[P_{\mathrm{OWF},i}^{\mathrm{rated}}]_1^{N_{\mathrm{OWF}}}, [P_{\mathrm{H},i}^{\mathrm{rated}}]_1^{N_{\mathrm{H}}},\\ & [S_{\mathrm{cable},i}]_1^{N_{\mathrm{cable}}}, [S_{\mathrm{BESS},i}]_1^{N_{\mathrm{BESS}}}\}_1^{N_0},\end{aligned} \tag{33}$$

$$\begin{aligned}\boldsymbol{Y}^0 =& \{\boldsymbol{y}_i\}_1^{N_0} = \{[f_1(\boldsymbol{x}_i)+\epsilon, f_2(\boldsymbol{x}_i), f_3(\boldsymbol{x}_i)]_i\}_1^{N_0}\\ =& \{[C_{\mathrm{ann}}^{\mathrm{ob}}(\boldsymbol{x}_i), -P_{\mathrm{RES}}(\boldsymbol{x}_i), C_{\mathrm{con}}(\boldsymbol{x}_i)]\}_1^{N_0}.\end{aligned} \tag{34}$$

Once the outcome set is firstly acquired or updated, the posterior distributions $\boldsymbol{p} = [p_i(f_i \mid \boldsymbol{D}, \sigma_{\mathrm{n},i})]_1^3$ are created using the Gaussian process over each objective function f_i.

$$f_i(\boldsymbol{x}) \sim p_i(f_i \mid \boldsymbol{D}, \sigma_{\mathrm{n},i}) = \mathcal{N}(\mu_{\boldsymbol{D},i}, \boldsymbol{\Sigma}_{\boldsymbol{D},i}), \quad i = 1,2,3, \tag{35}$$

with

$$\begin{cases}\mu_{\boldsymbol{D},i}(\boldsymbol{x}) = \boldsymbol{K}(\boldsymbol{x},\boldsymbol{X})[\boldsymbol{K}(\boldsymbol{X},\boldsymbol{X})+\sigma_{\mathrm{n},i}\mathbf{I}]^{-1}\hat{\boldsymbol{y}}_i\\ \boldsymbol{\Sigma}_{\boldsymbol{D},i}(\boldsymbol{x},\boldsymbol{x}') = \boldsymbol{K}(\boldsymbol{x},\boldsymbol{x}')\\ -\ \boldsymbol{K}(\boldsymbol{x},\boldsymbol{X})[\boldsymbol{K}(\boldsymbol{X},\boldsymbol{X})+\sigma_{\mathrm{n}}\mathbf{I}]^{-1}\boldsymbol{K}(\boldsymbol{X},\boldsymbol{x}')\\ \boldsymbol{K}(\boldsymbol{X},\boldsymbol{X}) = [k(\boldsymbol{x}_i,\boldsymbol{x}_j)]_{i,j=1}^n\\ = [\sigma^2 \dfrac{2^{1-\upsilon}}{\Gamma(\upsilon)}(\dfrac{\sqrt{2\upsilon}\left\|\boldsymbol{x}_i-\boldsymbol{x}_j\right\|}{\ell})^{\upsilon} K_{\upsilon}(\dfrac{\sqrt{2\upsilon}\left\|\boldsymbol{x}_i-\boldsymbol{x}_j\right\|}{\ell})]_{i,j=1}^n\end{cases}, \tag{36}$$

where $\mathcal{N}(\mu_{\boldsymbol{D},i}, \boldsymbol{\Sigma}_{\boldsymbol{D},i})$ is the generated multivariate normal distribution given data $\boldsymbol{D}$ with mean vector $\mu_{\boldsymbol{D},i}$ and covariance matrix $\boldsymbol{\Sigma}_{\boldsymbol{D},i}$; $\mathbf{I}$ is the identity matrix; $\hat{\boldsymbol{y}}_i$ represents the normalized version of $\boldsymbol{y}_i$; $k(\boldsymbol{x}_i, \boldsymbol{x}_j)$ is the Matérn kernel function with three kernel parameters σ, υ and l, all determined by maximum likelihood estimation; Γ is the gamma function, and K_υ is the modified Bessel function of the second kind.

4.2. Seeking of the candidate point

Once the posterior distributions are established, the acquisition function known as *noisy expected hypervolume improvement*, denoted as α_{NEHVI} is optimized to find the potential optimal capacity allocation schemes $\boldsymbol{x}_{\mathrm{cand}}$ as the candidate to be evaluated. The detailed process is demonstrated in Algorithm 1.

Algorithm 1 Locating the candidate to be evaluated

Require: $\boldsymbol{r}$: reference point; $\boldsymbol{p}$: posterior distributions; $\boldsymbol{D}$: evaluated outcome set.
Ensure: $\boldsymbol{x}_{\mathrm{cand}}$: candidate point.
1: **def** Pareto optimal set: $\mathcal{P}$
2: $\mathcal{P} = \{\boldsymbol{g}(\boldsymbol{x}) \text{ s.t. } \nexists\ \boldsymbol{x}' \in \boldsymbol{X} : \boldsymbol{g}(\boldsymbol{x}') < \boldsymbol{g}(\boldsymbol{x})\}$
3: **def** Hypervolume indicator: $HP(\mathcal{P} \mid r)$
4: $\boldsymbol{r} \in \mathbb{R}^M : HP(\mathcal{P} \mid \boldsymbol{r}) = \lambda_M \left(\bigcup_{\boldsymbol{v}\in\mathcal{P}}[\boldsymbol{r}, \boldsymbol{v}]\right)$
5: **def** Hypervolume improvement: $HVI(\mathcal{P}' \mid \mathcal{P}, r)$
6: $HVI(\mathcal{P}' \mid \mathcal{P}, r) = HP(\mathcal{P}' \cup \mathcal{P} \mid r) - HP(\mathcal{P} \mid r)$
7: **procedure** max $HVI(\boldsymbol{x}_{\mathrm{cand}})$
8: $\tilde{\boldsymbol{f}}_t \sim \boldsymbol{p}\left(\boldsymbol{f} \mid \boldsymbol{D}, \boldsymbol{\sigma}_{\mathrm{n}}\right)$, for $t = 1, \dots N_s$ ⊳ sample from $\boldsymbol{p}$
9: $\mathcal{P}_t \leftarrow \{\tilde{\boldsymbol{f}}_t(\boldsymbol{x}) \mid \boldsymbol{x} \in X_n, \tilde{\boldsymbol{f}}_t(\boldsymbol{x}) < \tilde{\boldsymbol{f}}_t(\boldsymbol{x}')\ \forall x' \in \boldsymbol{X}_n\}$
10: $\hat{\alpha}_{\mathrm{NEHVI}}(\boldsymbol{x}) \leftarrow \frac{1}{N}\sum_{t=1}^{N} HVI\left(\tilde{\boldsymbol{f}}_t(\boldsymbol{x}) \mid \mathcal{P}_t\right)$ ⊳ Approximate α_{NEHVI}
11: $\boldsymbol{x}_{\mathrm{cand}} \leftarrow \max\ \hat{\alpha}_{\mathrm{NEHVI}}(\boldsymbol{x})$
12: **end procedure**
⊳ The detailed calculation of line 4, 11 can be found in [33,34].

Algorithm 1 involves taking the average of multiple samples from the posterior distribution $\boldsymbol{p}$, which helps to neutralize the deviations. This ensures that the selection of potential optimal capacity allocation schemes is not misled by the noisy outcomes, avoiding a clumped Pareto front that lacks diversity.

4.3. Adaptive determination of the crucial parameters

The study builds upon the foundational work in [33,34] by refining the determination of two critical parameters: the reference point $\boldsymbol{r}$ used in Algorithm 1 and the noise standard deviation $\sigma_{\mathrm{n},1}$ in (36). Notably, the functions f_2 and f_3 are considered noiseless, indicating $\sigma_{\mathrm{n},2} = \sigma_{\mathrm{n},3} = 0$. While the original work assumes these parameters as known and fixed, this study presents an advancement by dynamically determining them, which enhances the algorithm's robustness and applicability.

4.3.1. Reference point

The selection of reference point $\boldsymbol{r}$ plays a significant part in calculating the hypervolume and thus affects the quality of the final Pareto front since hypervolume is the key metric to be optimized. Typically, a reference point marginally inferior to the worst-case scenario is deemed suitable for evaluating Pareto front quality. In [33], the reference point is a fixed value chosen by the decision-maker. However, due to the intricacies of the collaborative capacity planning model introduced in the study, pinpointing the worst-case scenario for each objective is a challenging task, and as stated before, a significant advantage of AMBO is its ability to eliminate predetermined values, thereby removing subjective bias from planners. Consequently, $\boldsymbol{r}$ is set to be an adaptive vector as detailed in (37):

$$\boldsymbol{r} = \hat{\boldsymbol{y}}^{\max} - d \cdot \hat{\boldsymbol{y}}^{\min}, \tag{37}$$

where $\hat{\boldsymbol{y}}^{\min}$, $\hat{\boldsymbol{y}}^{\max}$ are the minimum and maximum value of each column in $\hat{\boldsymbol{y}}$; $\hat{\boldsymbol{y}}^{\max}$ stands for the worst case that has been evaluated; d is the reference point adjustment factor and the minus of $d \cdot \hat{\boldsymbol{y}}^{\min}$ instead of a constant is to take the range of each objective into consideration.

4.3.2. Noise standard deviation

As for $\sigma_{\mathrm{n},1}$, the appropriate value of it is critical for the precision of the probabilistic surrogate model. $\sigma_{\mathrm{n},1}$ is affected by the penetration of RESs and other factors such as system topology, load levels, and more. These factors can vary a lot in different power systems. This makes setting a predetermined value for $\sigma_{\mathrm{n},1}$ a complex task that could require extensive and time-consuming experiments. To address this, the AMBO incorporates a marginal log likelihood based method, allowing $\sigma_{\mathrm{n},1}$ to self-update without prior assumptions. The specifics of this process are outlined in Algorithm 2.

Algorithm 2 Estimate noise standard deviation in the Gaussian process

Require: $\boldsymbol{x}$, $\boldsymbol{y}$: training data; n: number of data points; $k(\boldsymbol{x}_i, \boldsymbol{x}_j)$: kernel function; υ: tolerance; α: learning rate.
Ensure: σ_{n}: estimated noise standard deviation.

$\sigma_{\mathrm{n}} \leftarrow \mathrm{random}() > 0$ ▷ initialize with a positive number
repeat
  $\boldsymbol{K} \leftarrow [k(\boldsymbol{x}_i, \boldsymbol{x}_j)]_{i,j=1}^{n}$
  $\boldsymbol{K}_{\mathrm{n}} \leftarrow \boldsymbol{K} + \sigma^2 \mathbf{I}$
  $\mathrm{MLL} \leftarrow -\frac{1}{2}\boldsymbol{y}^T \boldsymbol{K}_{\mathrm{n}}^{-1} \boldsymbol{y} - \frac{1}{2}\log|\boldsymbol{K}_{\mathrm{n}}| - \frac{n}{2}\log(2\pi)$
  $\frac{\partial \mathrm{MLL}}{\partial \sigma_{\mathrm{n}}^2} \leftarrow \frac{1}{2}\,\mathrm{trace}\left(\boldsymbol{K}_{\mathrm{n}}^{-1} - \boldsymbol{K}_{\mathrm{n}}^{-1}\boldsymbol{y}\boldsymbol{y}^T\boldsymbol{K}_{\mathrm{n}}^{-1}\right)$
  $\sigma_{\mathrm{n}} \leftarrow \sqrt{\sigma_{\mathrm{n}}^2 + \alpha \frac{\partial \mathrm{MLL}}{\partial \sigma_{\mathrm{n}}^2}}$ ▷ update by gradient ascent
until $\|\nabla_{\sigma_{\mathrm{n}}^2}\mathrm{MLL}\| < \upsilon$ **or** $|\mathrm{MLL}_{\mathrm{new}} - \mathrm{MLL}_{\mathrm{old}}| < \upsilon$
return σ_{n}

Through iterative evaluation and inference, a broader range of non-dominant solutions can be uncovered. When finalizing the Pareto front, the objective function values $\boldsymbol{f}(\boldsymbol{x})$ are replaced with the means of posterior distributions $\boldsymbol{\mu}_{\boldsymbol{D}}^{J}$. For the $f_2(\boldsymbol{x})$ and $f_3(\boldsymbol{x})$, these two values are identical because they are noiseless. However, $f_1(\boldsymbol{x})$ is subject to noise, making the posterior mean a more accurate indicator when sufficient data is available compared to the simulation outcome carried out under a random RES scenario. The effectiveness of this substitution is validated in the next section.

4.4. Overall optimization process

The complete optimization process is illustrated in Algorithm 3.

4.5. Process of practical use

As illustrated in Fig. 4, in practice, the proposed approach is applied by first identifying suitable OWF locations [35] and designing the offshore-onshore connection topology, including key parameters such as wind profiles, load patterns, and network configurations. A specific simulation model based on the formulations in Section 3 is then built to represent the planned system's technical and operational characteristics. The AMBO algorithm iteratively generates and evaluates capacity allocation schemes for OWFs, offshore cables, hydrogen production systems, and BESS, producing a diverse Pareto front of planning solutions. Planners can then select schemes that align with specific interests such as economy, sustainability, or system reliability.

5. Case study

The case study is divided into three subsections. The simulation settings are introduced at first. In the second subsection, a set of typical scenarios, rather than a random RES scenario, is employed to evaluate the annual cost C_{ann}. This approach allows for a straightforward comparison of different MOOs. In the final subsection, the superiority of the proposed capacity planning model is demonstrated. The simulations are conducted on a computer with a 2.9 GHz Intel CORE i9-12900H processor and 32 GB of RAM. The sub-problems in the scheme evaluation model are solved by Gurobi.

Algorithm 3 Adaptive Multi-Objective Bayesian Optimization

Function AMBO
$j \leftarrow 0$, $\boldsymbol{D}^0 \leftarrow \{\boldsymbol{X}^0, \boldsymbol{Y}^0\}$ ▷ initialize
while $j < J$ **do** ▷ J: maximum number of iterations
  GP($\sigma_{\mathrm{n},1}^{j}$) **do**
    $p^j(\boldsymbol{f} \mid \boldsymbol{D}^j) \leftarrow \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{D}^j}, \boldsymbol{\Sigma}_{\boldsymbol{D}^j})$ ▷ generate distributions
    $\boldsymbol{r} = \hat{\boldsymbol{y}}^{\max} - \hat{\boldsymbol{y}}^{\min} * 10\%$ ▷ update reference point
  Algorithm 1 **do**
    $\boldsymbol{x}_{\mathrm{cand}} \leftarrow \max\ \hat{\alpha}_{\mathrm{NEHVI}}(\boldsymbol{x})$ ▷ find the candidate
  $\boldsymbol{y}_{\mathrm{cand}} \leftarrow \boldsymbol{f}(\boldsymbol{x}_{\mathrm{cand}}) + \epsilon$ ▷ evaluate the candidate
  $\boldsymbol{D}^{j+1} \leftarrow \boldsymbol{D}^j \cup \{\boldsymbol{x}_{\mathrm{cand}}, \boldsymbol{y}_{\mathrm{cand}}\}$ ▷ update observation set
  Algorithm 2 **do**
    $\sigma_{\mathrm{n},1}^{j+1} \leftarrow \sqrt{\sigma_{\mathrm{n}}^2 + \alpha \frac{\partial \mathrm{MLL}}{\partial \sigma_{\mathrm{n}}^2}}$ ▷ update noise std
  $j \leftarrow j + 1$
end while
GP($\sigma_{\mathrm{n},1}^{J}$) **do**
  $p^J(\boldsymbol{f} \mid \boldsymbol{D}^J) \leftarrow \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{D}}^{J}, \boldsymbol{\Sigma}_{\boldsymbol{D}}^{J})$
$\mathcal{P}^* \leftarrow \{\boldsymbol{\mu}_{\boldsymbol{D}}^{J}(\boldsymbol{x}^*)$ s.t. $\nexists\ \boldsymbol{x} \in \boldsymbol{X}^J : \boldsymbol{\mu}_{\boldsymbol{D}}^{J}(\boldsymbol{x}) < \boldsymbol{\mu}_{\boldsymbol{D}}^{J}(\boldsymbol{x}^*)\}$
return $\mathcal{P}^*$

5.1. Simulation settings

5.1.1. Parameters

The cost parameters in (5)–(8) are listed in Appendix A.1 with the corresponding sources.

5.1.2. Test system

The planning model utilizes a modified version of the IEEE 14-bus system, incorporating three additional OWFs as independent buses. The load is reallocated to illustrate the spatial mismatch between the load demand and offshore wind resources as in Fig. 5.

5.1.3. Algorithm

To address the challenges of noisy objectives and parameter sensitivity in power system planning, the AMBO algorithm is proposed with two key enhancements: robust noise handling through probabilistic modeling and adaptive determination of critical parameters. To validate these improvements, three representative algorithms are selected for comparison. NSGA-II, a widely used evolutionary algorithm, is included to demonstrate AMBO's superior sample efficiency, as AMBO requires significantly fewer evaluations by leveraging a surrogate model. PMBO [24] and NMBO [33], both Bayesian-based MOO methods, are chosen to highlight AMBO's advantage in adaptive parameter tuning and its enhanced robustness in the presence of uncertainty. Together, these comparisons illustrate the practical strengths of AMBO in solving complex, noisy, and computationally expensive planning problems.

5.1.4. Data

The load power curves for each bus are derived from the typical load curves of different provinces in China. RES scenarios are generated from wind speed data of different locations in the North Sea. The worst RES scenarios for the $N-1$ contingency analysis are those with the largest standard deviation in daily output power, as these represent the worst-case conditions for system stability.

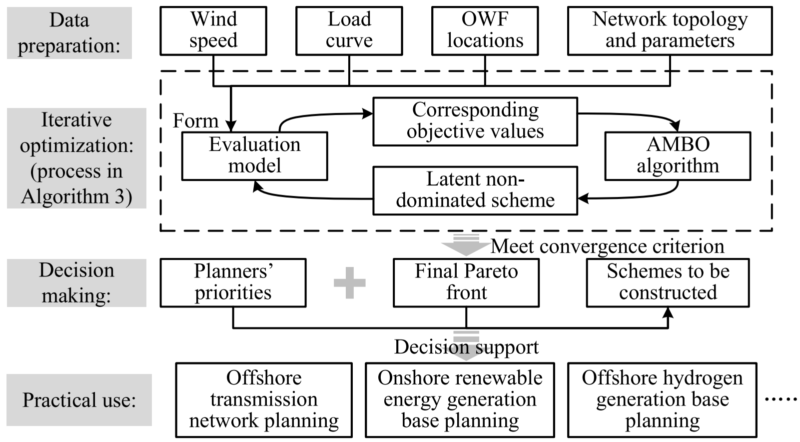


Fig. 4. The process of putting the proposed approach into practical use.

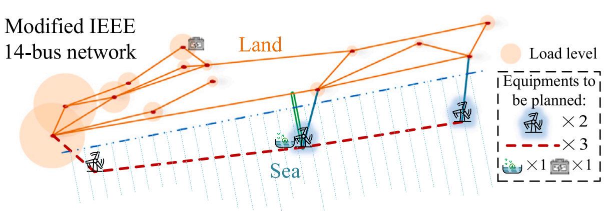


Fig. 5. The network used for simulation.

5.2. Effectiveness of the proposed approach

To straightforwardly demonstrate the advantages of AMBO and provide a more balanced validation, the annual cost is evaluated using a set of typical scenarios in this subsection, which renders all three objectives noiseless and allows for a fair comparison with conventional optimization methods that do not account for simulation noise.

The Pareto fronts generated by different algorithms are depicted in Fig. 6. From a visual perspective, the Pareto front generated by AMBO extends into regions of the objective space that other algorithms do not reach and exhibits a more uniform distribution of solutions. In contrast, NMBO and PMBO show localized clustering, limiting their exploration of the full solution space. Such limitations are also reflected in a quantitative perspective as shown in Fig. 7. The curves demonstrate that AMBO achieves a higher hypervolume than the other algorithms, indicating that it dominates a larger portion of the objective space and reflects superior convergence and coverage. Based on the comparisons, AMBO produces a Pareto front that spans a broader and more balanced region of the objective space, reflecting a richer diversity of non-dominated solutions. Moreover, NSGA-II evaluates over ten times more points but achieves the lowest hypervolume, underscoring the sample-efficiency of Bayesian-based MOO algorithms. It is important to note that the hypervolume curves represent average outcomes from multiple optimization runs, and a reference point slightly inferior to the worst-case scenario among all evaluated points is chosen to ensure unbiased assessment.

5.3. Superiority of the proposed approach

In this subsection, the typical scenarios are abandoned and Algorithm 3 is completely executed.

As highlighted earlier, all capacity allocation schemes are designed to be operational in a future that is inherently uncertain. Consequently, the actual annual cost $C_{\text{ann}}(\boldsymbol{x})$ of these schemes is difficult to assess precisely through simulation alone. The study employs the sample average approximation (SAA) [36] to establish a benchmark that closely approximates the real annual cost. Each scheme undergoes multiple evaluations under various RES scenarios and the mean value $C_{\text{ann}}^{\text{SAA}}$ obtained is considered the actual annual cost of the scheme in question. Detailed formulations are shown below:

$$C_{\text{ann}}(\boldsymbol{x}) \approx C_{\text{ann}}^{\text{SAA}}(\boldsymbol{x}) = \frac{1}{N_{\text{sce}}} \sum_{i=1}^{N_{\text{sce}}} C_{\text{ann}}^{\text{ob}}(\boldsymbol{x} \mid P_{\text{res},i}), \tag{38}$$

$$\bar{e} = \frac{1}{N_{\text{sch}}} \sum_{i}^{N_{\text{sch}}} \frac{\mu_{\boldsymbol{D},1}(\boldsymbol{x}) - C_{\text{ann}}^{\text{SAA}}(\boldsymbol{x})}{C_{\text{ann}}^{\text{SAA}}(\boldsymbol{x})} \times 100\%, \tag{39}$$

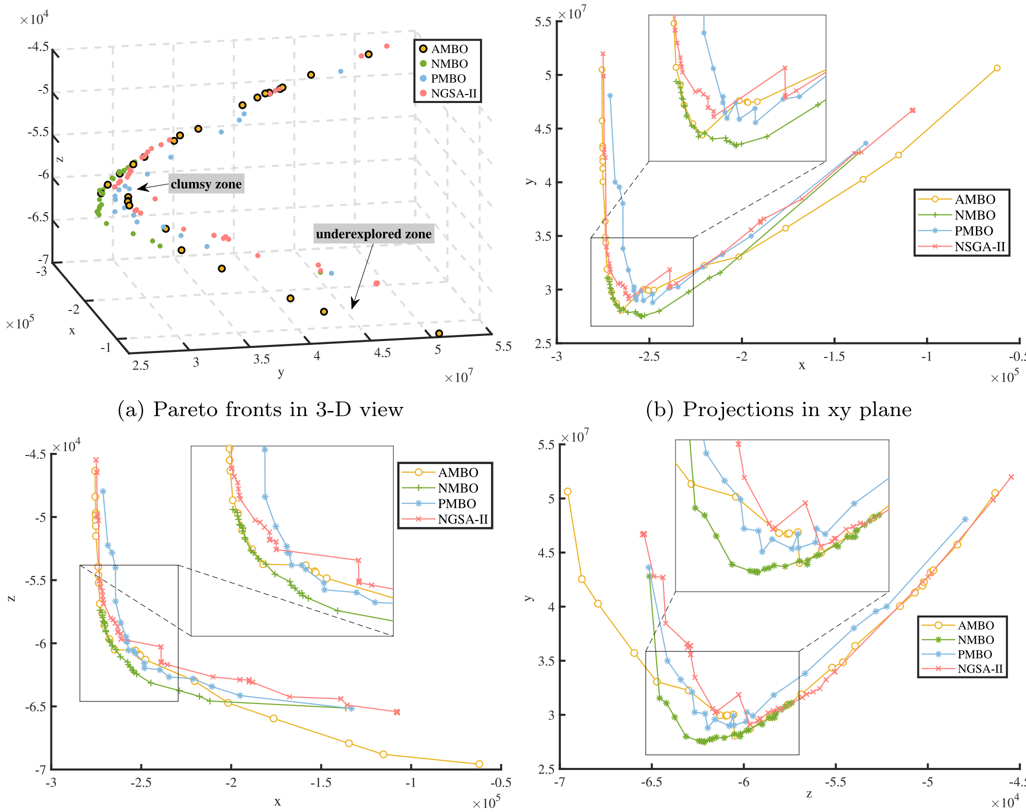
where N_{sce} is the number of scenarios used to evaluate each scheme; $\bar{e}$ is the average error to evaluate the inference accuracy. N_{sch} is the number of evaluated schemes; $\mu_{\boldsymbol{D},1}(\boldsymbol{x})$ is the Gaussian process mean acquired in (36).

The comparison in Fig. 8 verifies that the proposed AMBO framework maintains a consistently low inference error, about 3% even under varying noise levels caused by RES variability. This capability arises from its adaptive estimation of $\sigma_{\text{n},1}$, allowing the algorithm to self-adjust to fluctuating renewable outputs without prior parameter tuning. In contrast, NMBO achieves similar accuracy only when $\sigma_{\text{n},1}$ is predetermined with precision, which is both computationally expensive and sensitive to specific system models.

The results of mean error testified that AMBO achieves high evaluation accuracy without compromising either objectivity or computational efficiency. By embedding a noise-modeling mechanism directly into the optimization and combining it with a probabilistic surrogate model, AMBO inherently accommodates the inevitable deviation between simulation and real-world operation. This ensures that the optimization process remains stable and delivers diverse, high-quality Pareto fronts even under highly variable RES conditions, demonstrating strong robustness for practical capacity planning.

The hypervolume indicator curves presented in Fig. 9 are derived from the average values of multiple independent optimization runs to ensure the reliability of performance comparisons. The figure highlights a pronounced performance gap between NSGA-II and the Bayesian-based MOO algorithms, as NSGA-II lacks the capacity to filter out noise, often resulting in misleading convergence behavior. AMBO consistently demonstrates strong performance, maintaining a clear advantage in convergence speed and stability.

Table 1 further compares the four algorithms across several key metrics. Although NMBO generates a slightly higher number of PF points on average, these solutions are densely clustered in certain regions of the objective space, limiting their contribution to the diversity of the Pareto front and resulting in a lower final hypervolume. In contrast, AMBO not only provides a competitive number of PF points but also ensures



(a) Pareto fronts in 3-D view

(b) Projections in xy plane

(c) Projections in xz plane

(d) Projections in yz plane

Fig. 6. Visualization of the Pareto fronts derived from the four optimization algorithms in the three-objective planning problem (x-axis: negative renewable energy consumption $-P_{\mathrm{RES}}$ (MW), y-axis: annual cost C_{ann} (M€), z-axis: reserved power-to-investment ratio $-RIRP$ (W/€)). Subfigure (a) shows the complete 3D Pareto fronts with discrete points representing non-dominated solutions. Subfigures (b), (c), and (d) present the 2D projections of these Pareto fronts onto the xy, xz, and yz planes, respectively. In these projections, lines connect the Pareto front points to highlight the relative spread and shape of each front.

Table 1
Comparison of performance metrics across the four MOO algorithms.

Algorithms	Hypervolume indicator	Number of PF points	Generational distance	p-value from Wilcoxon signed-rank test
AMBO	85.79	29.80	1.5633	–
NMBO	78.19	31.75	2.2475	0.1611
PMBO	76.95	23.33	1.9426	0.0003
NSGAII	60.82	4.67	2.7845	0.0000

The details of calculating the metrics can be found in Appendix A.2

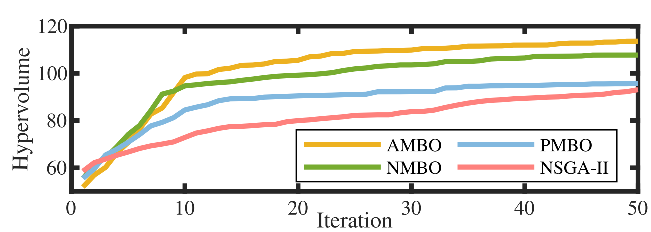


Fig. 7. Progress of the hypervolume indicator for the four algorithms under noiseless scenarios, using a set of typical RES scenarios.

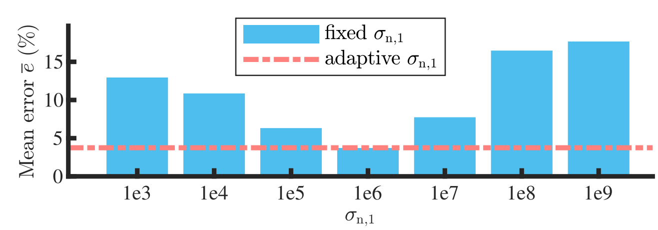


Fig. 8. Comparison of the mean error between NMBO and AMBO across varying noise standard deviation $\sigma_{\mathrm{n},1}$. Bars represent the average error of NMBO with different predetermined $\sigma_{\mathrm{n},1}$, while the horizontal dashed line indicates the corresponding average error of AMBO. The error reflects the mean difference between the predicted values from the final Gaussian process model and the test values obtained via sample average approximation.

a broader and more balanced distribution. It also achieves the lowest generational distance among all algorithms, indicating its solutions are closest, on average, to the reference Pareto front composed of the best solutions found across all runs, highlighting superior convergence capability.

According to the Wilcoxon test [37] performed on the final hypervolume indicator, AMBO significantly outperforms PMBO and NSGA-II, while the result against NMBO ($p = 0.1611$) indicates a less statistically significant difference. Nevertheless, it is worth noting that the performance of NMBO and PMBO hinges on time-consuming parameter

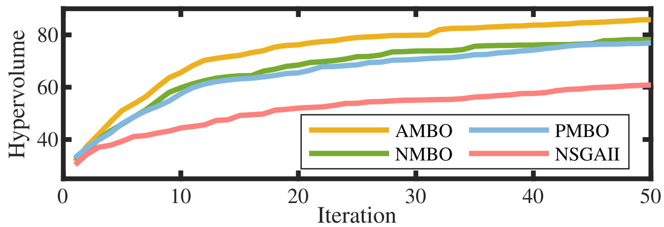


Fig. 9. Progress of the hypervolume indicator for the four algorithms under a noisy circumstance, where each scheme's performance is evaluated with a randomly picked RES scenario.

Table 2
The settings of four capacity planning models.

Model	OWFs	Offshore cables	Hydrogen electrolyzer	BESS
0	✓	–	–	–
1	✓	–	–	✓
2	✓	✓	–	✓
3	✓	✓	✓	✓

[a] In Model 0 & 1, the OWFs are connected to land in a point-to-point way by cables having a capacity that is 90% of the maximum capacity of their corresponding OWFs.

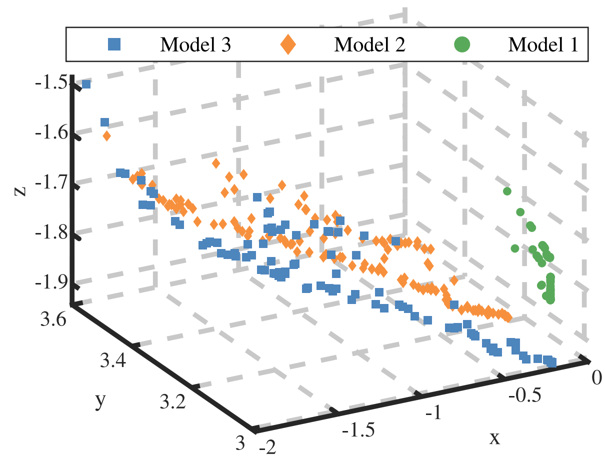


Fig. 10. The Pareto fronts acquired by AMBO in different models. (x: $-P_{\mathrm{RES}}$ (MW), y: C_{ann} (M€), z: $RIRP$ (W/€)).

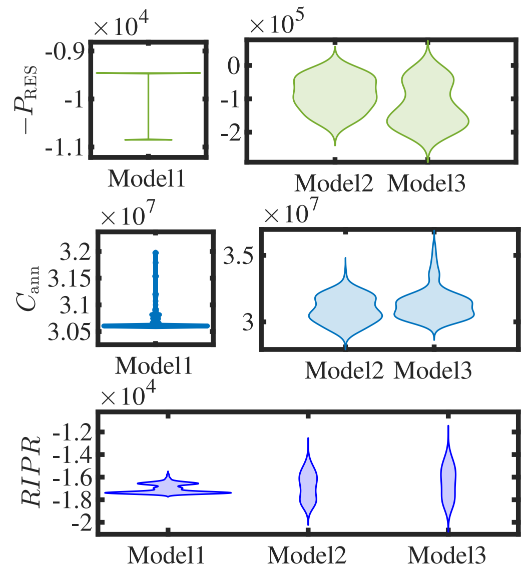


Fig. 11. The objective value distributions of the non-dominant solutions of the three models.

tuning, whereas AMBO achieves its results through an adaptive parameter selection mechanism. This not only enhances its optimization efficiency but also improves practical robustness. Overall, the results affirm AMBO's strength in balancing convergence quality, solution diversity, and ease of application, making it a compelling option for complex multi-objective planning tasks.

To demonstrate the benefits of connecting OWFs and integrating hydrogen electrolyzers, models with different equipment to be planned are outlined in Table 2. The Pareto fronts obtained by AMBO for these models are shown in Fig. 10. It is important to note that Model 0 is not feasible under certain RES scenarios, resulting in no attainable Pareto front.

The comparison between Model 2 and Model 3 with Model 1 in Fig. 10 shows that without offshore cable interconnections, added offshore wind capacity does not translate into higher renewable energy consumption due to transmission congestion. The non-dominated solutions from Model 3 generally reflect lower annual costs and higher renewable usage, supported by the integration of a hydrogen electrolyzer that converts surplus power into green hydrogen, enhancing overall efficiency. To evaluate the overall quality of solutions, the hypervolume indicator is used. Model 3 achieves the highest hypervolume (85.79), outperforming Model 2 (74.37) and Model 1 (4.59), indicating broader and more favorable trade-offs. Additionally, a violin plot as in Fig. 11 shows that Model 3 spans the widest range across all objectives, and its solutions exhibit, on average, an 8% decrease in levelized cost of renewable energy, reflecting stronger cost-effectiveness and planning flexibility.

These results highlight the expanded and more diverse solution space enabled by hydrogen integration and offshore interconnection, offering planners a wider set of high-quality trade-offs.

5.4. Sensitive analysis

5.4.1. Investment cost

Owing to anticipated advancements in technology, manufacturing efficiency, and broader industry developments, the investment cost associated with the proposed offshore power system is projected to decrease over the coming decades. To evaluate the impact of such cost reductions, a sensitivity analysis is conducted using a cost reduction factor c, which quantifies the proportion of cost decrease. The formulation of the reduced cost is provided in (40) and the results are illustrated in Figs. 12 and 13.

$$C'_{\mathrm{ann}} = c \sum_{k\in\Omega} CPR_k C_k + \sum_{k\in\Omega} \kappa_k C_k + (C_{\mathrm{g}} - R_{\mathrm{H}}), \tag{40}$$

As the investment cost declines, the mean values of consumed RES power remain relatively stable due to the constraint imposed by the total load level. However, the violin plots in Fig. 12 reveal that the distribution of consumed RES power becomes more uniform across the Pareto front, suggesting that reduced investment costs enhance planning flexibility by enabling a more balanced set of trade-offs across objectives.

This improvement in planning flexibility is particularly relevant in real-world settings, where system planners typically operate under implicit budget constraints even if these are not explicitly modeled. Under such constraints, lower infrastructure costs allow planners to achieve the same level of RES integration with reduced total investment, thereby freeing up financial resources for other priorities. This expanded allocation capacity enables a wider range of design choices, such as enhancing operational reliability, integrating carbon capture technologies, or deploying advanced voltage support equipment like

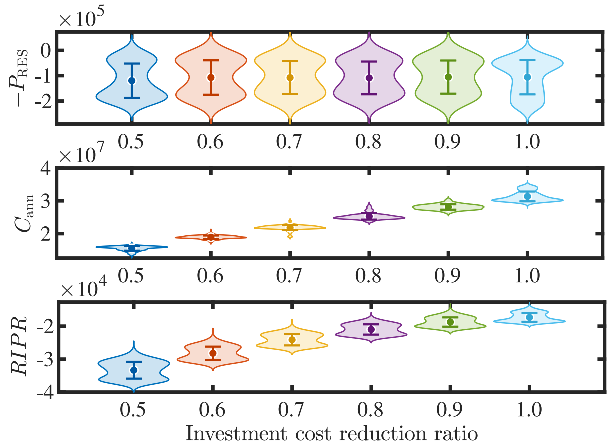


Fig. 12. Distribution of Pareto-optimal objective values under varying investment cost reduction ratios, with violin plots showing the empirical distributions at each cost reduction level and dots along with error bars representing the mean and standard deviation, respectively.

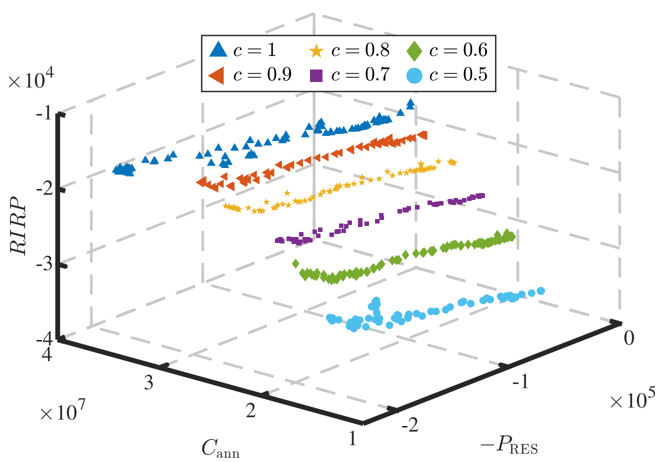


Fig. 13. The Pareto fonts acquired under different investment cost reduction ratios c.

VSCs, which, although outside the scope of the present model, illustrate how cost reductions can broaden the feasible design space and support more comprehensive, long-term planning strategies.

From a system security perspective, the metric $RIPR$ exhibits an upward trend in its mean value as the investment cost decreases. This implies that, under the same investment level, a more significant improvement in system robustness against $N-1$ contingencies can be achieved. Moreover, the distribution of $RIPR$ values also broadens, suggesting increased variability and a wider range of feasible solutions as costs decline. In summary, reductions in investment cost contribute not only to enhanced system security but also provide planners with a broader and more diversified set of Pareto optimal solutions. This enables more flexible decision-making, accommodating a variety of planning objectives such as operational reliability, cost efficiency, and RES integration.

5.4.2. BESS power-to-energy ratio

To investigate the influence of the BESS power-to-energy ratio on system planning outcomes, a sensitivity analysis was conducted across five representative ratios. The results, illustrated in Fig. 14, combine a bar chart and beeswarm plot to depict the distribution and mean values of the BESS capacities among Pareto-optimal solutions under each tested ratio.

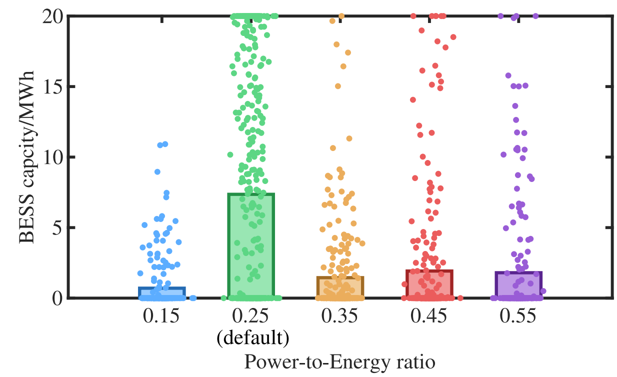


Fig. 14. Comparison of BESS capacities in Pareto-optimal planning solutions across varying power-to-energy ratios. Bars represent the mean planned BESS capacity at each ratio, with the beeswarm dots illustrating the distribution of individual Pareto-optimal solutions. .

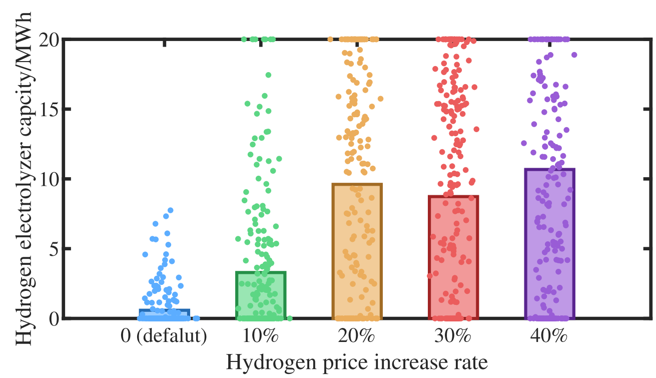


Fig. 15. Comparison of hydrogen electrolyzer capacities in Pareto-optimal planning solutions across varying power-to-energy ratios. Bars represent the mean planned electrolyzer capacity at each ratio, with the beeswarm dots illustrating the distribution of individual Pareto-optimal solutions.

As illustrated in the figure, a lower power-to-energy ratio (15%) leads to a significantly reduced BESS capacity in the planning results. This is primarily because the limited power output restricts the BESS's ability to effectively respond to net load fluctuations, leaving a substantial portion of its energy storage capacity underutilized. In contrast, the default 25% ratio enables a more balanced utilization of BESS resources. At this setting, multiple Pareto-optimal solutions reach the upper limit of allowable installed capacity, suggesting that the system's demand for power flexibility is not fully met under this ratio.

This limitation is notably alleviated when the ratio is increased to 35%. At this level, most BESS capacities in the Pareto-optimal solutions fall below half of the maximum allowable capacity. This implies that the system's flexibility requirements can be satisfied with smaller energy storage sizes when higher power output is available. Since enhancing the BESS's power output capability typically incurs lower costs than expanding energy capacity, a 35% ratio offers a more economically viable configuration. However, further increases in the ratio to 45% and 55% do not yield substantial additional improvements in system performance. These higher ratios result in only marginal cost increases without providing significant gains in operational flexibility or renewable energy integration.

In summary, while the default ratio of 25% aligns with current engineering practices, the sensitivity analysis suggests that adopting a higher ratio, such as 35%, can better address the power fluctuations from offshore wind farms and lead to a more cost-effective and technically efficient system configuration.

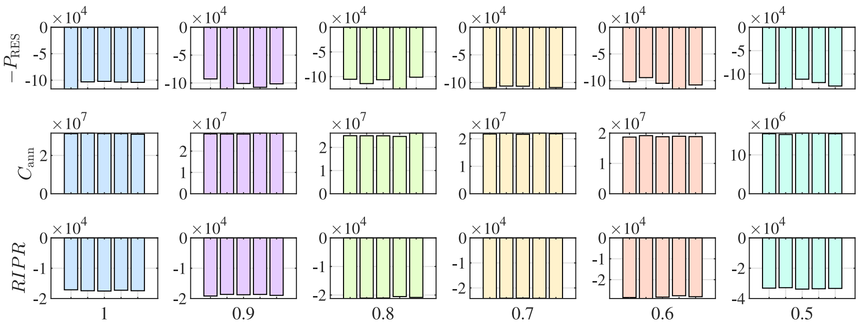


Fig. 16. Mean values of three objectives, P_{RES}, C_{ann}, and $RIPR$, achieved by the AMBO algorithm under six cost reduction ratios: 1.0, 0.9, 0.8, 0.7, 0.6, and 0.5 (left to right). Each column of subplots corresponds to one cost reduction ratio, and each bar within a subplot represents the result of an independent optimization run. The results show that, despite the inherent randomness of AMBO, the optimization outcomes remain consistent across repeated runs and varying cost reduction ratios.

5.4.3. Green hydrogen price

With the advancement of industrial decarbonization efforts and increasing global emphasis on clean energy transitions, green hydrogen is expected to become more economically attractive in the future. To examine how this potential price shift would influence the deployment of offshore hydrogen production systems, a sensitivity analysis was performed by increasing the unit price of green hydrogen from 1.1 to 1.4 times its baseline value, corresponding to a hydrogen price increase rate of 10% to 40%.

The results are illustrated in Fig. 15, which adopts a combination of bar chart and beeswarm plot to visualize the distribution and average values of hydrogen electrolyzer capacities in the Pareto-optimal solutions under each price level. At the current baseline price, offshore hydrogen production is generally not economically competitive, as most Pareto-optimal planning schemes select zero or minimal electrolyzer capacity. This outcome highlights the economic barrier to large-scale offshore hydrogen development under existing market conditions.

However, a small price increase leads to a dramatic shift. With only a 10% price increase, the average hydrogen electrolyzer capacity across Pareto-optimal solutions grows to nearly six times that of the baseline scenario. A further increase to 20% brings an additional threefold rise in capacity. These sharp increases indicate a high sensitivity of hydrogen investment decisions to small price changes, particularly when there is available surplus power from offshore wind farms. That said, increasing the hydrogen price beyond 20 results in diminishing returns. The average hydrogen electrolyzer capacity plateaus at around 9.8 MW under a 20% price increase, as this level already utilizes the majority of surplus offshore wind power. Further expansion of hydrogen capacity would require dedicated offshore wind farm investment solely for hydrogen production, despite the fact that power system load demand has already been met. This additional investment provides little economic benefit and thus is rarely selected in optimal solutions.

The increased electrolyzer capacity leads to measurable improvements in system-wide performance. Compared with the default case, the total consumed renewable energy sees an average increase of approximately 13%, indicating better utilization of offshore wind power. The levelized cost of renewable electricity decreases by 16%, reflecting enhanced economic efficiency. Furthermore, when compared to Model 2 in Table 2, which excludes offshore hydrogen production, the 20% hydrogen price increase scenario results in a 20% rise in RES consumption and a 21% decrease in levelized cost. These quantitative outcomes highlight the substantial economic and energy benefits enabled by integrating hydrogen production under more favorable pricing assumptions.

These findings suggest that even moderate improvements in green hydrogen market prices can effectively activate the deployment potential of offshore hydrogen systems. The proposed planning framework demonstrates clear responsiveness to such changes, enabling planners to understand both the threshold conditions and scale of economically justified hydrogen investment. This adaptability reinforces the value of the framework as a strategic tool for supporting data-driven decisions in future offshore energy planning.

5.5. Stability of optimization results

Since the proposed AMBO algorithm is built upon the classic Bayesian optimization framework, it inherently involves randomness in both the initialization and the iterative optimization processes. As such, even under identical parameter settings, slight variations in results could occur due to stochastic sampling and surrogate model updates. To verify that these random elements do not compromise the reliability of the final planning solutions, this subsection evaluates the stability of the results under fixed system conditions.

A case study was selected in which only the investment cost reduction ratio was varied, used as an illustrative example. For each ratio, the algorithm was executed over five independent runs with identical parameter settings to isolate the influence of stochastic elements from changes in external assumptions.

As shown in Fig. 16, despite the probabilistic sampling and surrogate modeling inherent in AMBO, the optimization consistently converged toward mean values of the objective values, exhibited minimal variation across repeated runs for the same cost reduction ratio.

This consistency demonstrates that the AMBO framework reliably produces consistent trade-off solutions when system parameters are held constant, reinforcing its suitability for practical planning applications where reproducibility is critical.

6. Conclusion

In the study, the proposed AMBO algorithm is verified to outperform traditional MOO algorithms in terms of optimization efficiency and the diversity of the final Pareto front. This enables planners to select capacity allocation schemes that better accommodate renewable energy (RES) and enhance system security within reasonable investment limits. By incorporating a surrogate probabilistic model, the AMBO algorithm also effectively mitigates the side effects of simulation deviation caused by renewable variability, without requiring excessive computational resources.

Moreover, the case study reveals the practical significance of the proposed approach in real-world offshore system planning. The interconnection of OWFs proves effective in enhancing the consumption of RES and improving power system stability. The integration of hydrogen production further boosts investment efficiency by converting surplus wind power into green hydrogen, thereby reducing curtailment and enabling new revenue streams. The approach offers a scalable, noise-aware framework that is aligned with typical planning needs, from

Table 3
Parameters Related to the Cost Calculation.

Equipment	Parameter	Value	Parameter	Value
OWF [28]	α_{OWF}	1444.074	β_{OWF}	381.513
	γ_{OWF}	0.87	δ_{OWF}	1.06
	α_{plat}	457.110	β_{plat}	56 013.675
	α_{WFconv}	77.630	T_{OWF}	25
Cable [28]	α_{cable}	15.807	β_{cable}	801.940
	γ_{cable}	1.16	T_{cable}	25
Hydrogen [29]	α_{H}	845.5	β_{H}	2.0978
	α_{pipe}	2.0978	α_{Hconv}	38.297
	T_{H}	25	η_{H}	50%
	$\mathrm{LHV_H}$	33	p_{H}	6 €/kg
BESS [30]	α_{BESS}	1	T_{BESS}	15
	λ_c	0.9	λ_d	0.9
Traditional generator	a_{1-5}	3/3/3/3/3	b_{1-5}	0.043/0.25/0.01/0.01/0.01
	c_{1-5}	20/20/40/40/40		

[a] The units for all the cost values and capacities are k€$_{2021}$ and MW/MWh by default.

Table 4
Parameters related to the simulation model in case study.

Type	Parameter	Value	Parameter	Value
Number of equipment	N_{OWF}	2	N_{cable}	3
	N_{BESS}	1	N_{H}	1
NSGA-II parameters	Population size	12	Tournament size	2
Cable/Pipe distance	$L_{\mathrm{cable},1}$	13.20 km	$L_{\mathrm{cable},1}$	151.39 km
	$L_{\mathrm{cable},3}$	145.96 km	$L_{\mathrm{pipe},1}$	42.7 km
Traditional generator	$P^{\max}_{g,1-5}$	664.8/280/200/200/200	δ_g	0.6
BESS	s	25% (default)	$S_i^{\min}$	$5\% * S^{\max}_{\mathrm{BESS},i}$
VCL	υ_1	10%	υ_2	5%
Wind speed data		Historical data in the North Sea		
Load data		Historical data in provinces of China		

offshore network layout to multi-scenario evaluation, and provides a diverse set of environmental friendly, cost-effective planning options. These characteristics make it a valuable decision-support tool for stakeholders, system planners, and policymakers pursuing long-term offshore renewable development under complex and uncertain conditions.

Despite its demonstrated advantages, the proposed AMBO algorithm presents some limitations that must be acknowledged. Its optimization efficiency tends to decline when dealing with high-dimensional decision spaces, highlighting the need for more scalable strategies. Furthermore, the current framework incorporates random sampling processes, which can render the outcomes irreproducible in some cases. Future research is being conducted to address these limitations and improve the practicality and robustness of the proposed approach in large-scale power system planning problems.

CRediT authorship contribution statement

Ruizhe Yang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Ying Xu:** Writing – review & editing, Supervision, Methodology. **Zhongkai Yi:** Writing – review & editing, Visualization, Validation, Supervision, Methodology. **Zhimin Li:** Writing – review & editing, Supervision. **Zhenghong Tu:** Writing – review & editing. **Junfei Wu:** Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

A.1. The parameters used in the planning model

The parameters concerning the investment calculation in (2)–(8) are listed in Table 3 with their corresponding sources, and the parameters related to the simulation model in Section 5 are listed in Table 4.

A.2. Details of calculating of metrics in Table 1

All metrics are computed based on multiple independent runs of each optimization algorithm.

Number of PF Points: The number of PF points is calculated by first identifying the unified PF set from all algorithms, then counting each algorithm's unique contributions to this global set, averaged over its total runs.

Generational distance: Since the true Pareto front is not available, the generational distance is calculated using the best non-dominated points gathered collectively from all four algorithms as an approximation.

Wilcoxon signed-rank test: The statistical comparison is conducted on the final hypervolume indicators obtained from multiple independent runs.

Table 5
Hypervolume results from AMBO algorithm with various reference point adjustment factors in multiple test problems.

d =	4%	8%	10% (default)	12%	16%	20%
Proposed model	81.35	85.49	85.79	86.02	84.44	80.79
BraninCurrin problem	51.35	58.27	58.35	58.22	58.27	55.40
ZDT1 problem (dim = 8)	117.09	118.65	120.22	119.75	118.90	118.11
DTLZ1 problem (dim = 6)	96 765.42	151 868.50	153,422.71	158 089.53	155240.18	155 240.18

Each test problems are assigned with a fixed reference point only used to calculate the hypervolume indicator.

A.3. The justification of the reference adjustment factor

A sensitivity analysis was conducted to evaluate the impact of the reference point adjustment factor d in Eq. (37) on the hypervolume indicator. The factor d was tested at values of 4%, 8%, 10%, 12%, 16%, and 20% across the proposed offshore planning model and three benchmark problems: Branin-Currin, ZDT1, and DTLZ1. Table 5 summarizes the hypervolume values obtained for each setting.

The results show that the 10% adjustment factor consistently delivers optimal or near-optimal hypervolume performance in most cases, with only a slight improvement at 12% for the DTLZ1 problem. Hypervolume values remain stable within the 8% to 15% range, indicating the method's robustness and low sensitivity to moderate changes in d. Given the normalization of objectives, this scale-invariant adjustment factor provides a reliable and practical reference point for hypervolume calculation in diverse problem settings.

Data availability

Data will be made available on request.

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